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ROTARY OPERATING DEVICE AND ELECTRONIC DEVICE

Abstract

A rotary operating device includes a rotary operating member **20** that can rotate around a rotation axis, a reflecting plate **70** that can rotate integrally with the rotary operating member, and a detection unit **95** that detects reflected light from the reflecting plate **70**. The reflecting plate **70** has a plurality of patterns as a detection target portion. A first pattern **70a1** is a circular portion that switches between a bright portion and a dark portion every 180 degrees around the rotation axis. A second pattern **70a2** is a circular portion that switches between a bright portion and a dark portion every 90 degrees around the rotation axis. An angle at which the first pattern **70a1** switches between the bright portion and the dark portion coincides with the angle at which the second pattern **70a2** switches between the bright portion and the dark portion.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] This invention relates to a rotary operating device, such as a dial device, and an electronic device equipped with a rotary operating device.

Description of the Related Art

[0002] Dial devices with which electronic devices such as digital cameras, video cameras, and portable information terminals are equipped are used for changing operating modes and various settings. For example, a dial device has a conductive pattern fixed to the electronic device main body to detect the rotation phase, and consists of a plate spring-like phase contact piece that slides against the conductive pattern and is fixed to a rotatable dial.

[0003] In addition, a dial device has a plurality of grooves arranged side by side in the direction of rotation and a sphere that engages with each groove, and is configured to engage and disengage the sphere and grooves as the dial is rotated, generating a clicking sensation in the rotary operation of the dial.

[0004] In a configuration for using sliding between the phase contact strip and the conductive pattern, in a case where wear particles are generated, they may cause unintended shorts between signals, resulting in false detection. Japanese Patent Laid-Open No. 2022-191824 discloses a rotary switch for detecting the phase of the rotary operating member by photo-interrupters and slits in a non-contact manner.

[0005] The rotary switch has a detection target portion formed integrally with the operating axis, and is configured to detect the presence or absence of a slit formed on a rotary position detectable body or a status detectable body by a plurality of multiple photo-interrupters. Any one of $2n$ states, which is the number of combinations of ON/OFF of the detection outputs of the n photo-interrupters, is specified.

[0006] Incidentally, the rotation angle (resolution) of the dial device is set at a desired number of phases, and the shape of the conductive pattern differs for each phase number. Therefore, in a case where a different number of phases is to be set, the shape of the conductive pattern needs to be changed each time.

[0007] In addition, in the conventional technology disclosed in Japanese Patent Laid-Open No. 2022-191824, the number of inner slits is provided according to the desired state, which may cause the device to become larger. In addition, phase detection is performed by the detection signals at the rotary position detectable body or the status detectable body, and the presence or absence of a slit in the status detectable body is detected.

[0008] However, the position of the slit is fixed, and the combination of the detection output of the photo-interrupter pertaining to the status detectable body is limited. Therefore, the rotary operating member is configured for reciprocating rotation and cannot be rotated all the way around.

SUMMARY OF THE INVENTION

[0009] The rotary operation device of the embodiment of the present invention includes a rotary operating member that can rotate around a rotary axis, a reflective member that can rotate together with the rotary operating member, and a detection unit configured to detect reflected light from the reflective member, in which the reflective member is provided with a plurality of detection target portions each having a bright portion and a dark portion. A first detection target portion of the plurality of detection target portions is a portion formed in an annular shape in which a bright portion and a dark portion switch every 180 degrees around the rotation axis, A second detection target portion of the plurality of detection target portions is a portion formed in an annular shape in which a bright portion and a dark portion switch every 90 degrees around the rotation axis. An

angle at which the bright portion and dark portion switch at the first detection target portion is the same as the angle at which the bright portion and dark portion switch at the second detection target portion.

[0010] Further features of the present invention will become apparent from the following description of embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1 is a perspective view illustrating a main body of a camera of an embodiment.

[0012] FIGS. 2A and 2B are exploded perspective views illustrating a configuration of a rotary operating device according to the embodiment.

[0013] FIG. 3 is a cross-sectional view illustrating the configuration of the rotary operating device according to the embodiment.

[0014] FIGS. 4A to 4C are diagrams for illustrating an arrangement of a reflecting plate and a detection unit in the embodiment.

[0015] FIGS. 5A-1 to 5E-2 are diagrams for illustrating a detection target portion of the reflecting plate and a detection signal for each phase number.

[0016] FIGS. 6A to 6G are diagrams for illustrating an example of a rotary operating member having six phases.

[0017] FIGS. 7A and 7B are perspective views illustrating another example of a retention mechanism that generates a clicking sensation.

[0018] FIG. 8 is a perspective view illustrating still another example of a retention mechanism that generates a clicking sensation.

DESCRIPTION OF THE EMBODIMENTS

[0019] Hereinafter, with reference to the accompanying drawings, favorable modes of the present invention will be described using Embodiments. In each diagram, the same reference signs are applied to the same members or elements, and duplicate description will be omitted or simplified.

[0020] In the embodiment, an imaging device will be described as an example of an electronic device equipped with a rotary operating device. Referring to FIGS. 1 to 3, the configuration of the rotary operation module for the embodiment is described.

[0021] FIG. 1 is a perspective view illustrating a main body of a digital camera (hereinafter, simply referred to as the “camera”) 1. For example, the camera 1 is a digital single-lens reflex camera. The main body of the camera 1 is equipped with a mode dial unit 2 and an exterior cover 3. In the following description, the side of the main body of the camera 1 on which the exterior cover 3 is provided is defined as the upper side.

[0022] The mode dial unit 2 is provided on the top surface of the main body of the camera 1. The mode dial unit 2 is a rotary operating device used by the user to operate and set various imaging conditions.

[0023] For example, the user can select the desired imaging mode by rotating the mode dial unit 2 in a clockwise or counterclockwise direction. Imaging modes include a shutter speed priority mode (Tv mode), an aperture priority mode (Av mode), and a movie mode.

[0024] Referring to FIGS. 2 and 3, the rotary operation of the mode dial unit 2 will be described.

FIG. 2A is an exploded perspective view of the mode dial unit 2 as viewed obliquely from below.

FIG. 2B is an exploded perspective view of the mode dial unit 2 as viewed obliquely from below.

FIG. 3 is a cross-sectional view of the mode dial unit 2. The rotary operating member 20 of the mode dial unit 2 is a member that is rotated by the user and can be rotated 360 degrees. The mode dial unit 2 is equipped with a retention mechanism (hereinafter, referred to as “click mechanism”) to generate a clicking operation force (click force) in accordance with the rotary operation of the

rotary operating member **20**.

[0025] The click mechanism has a sphere, an elastic member, and a groove. Click balls **30** and **31** are arranged in grooves **21** provided inside the rotary operating member **20**, and click forces can be generated by biasing the click balls **30** and **31** in the direction of the rotation axis by compression springs **40** and **41**.

[0026] The compression springs **40** and **41** and the click balls **30** and **31** are assembled into the holes of storage portions **51** and **52** in an exterior member **50**. The click mechanism will be described in detail later.

[0027] The rotary operating member **20** is rotatably supported by the exterior member **50** with its shaft portion **22** slidably fitted into a hole portion **53** of the exterior member **50**. A convex **23** on a shaft **22** is fixed in rotation phase to the hole **61** on a receiving plate **60** of a reflecting plate **70** and is fastened with screws **80**.

[0028] The reflecting plate **70** is a reflective member having a bright portion where the reflectance is equal to or greater than a threshold value and a dark portion where the reflectance is less than the threshold value. The reflecting plate **70** is positioned by convexes **62**, **63**, and **64** on the receiving plate **60** and concavities **71**, **72**, and **73** on the reflecting plate **70**, and is fixed by adhesion using double-sided adhesive tape, or the like, not illustrated. A fixing member **100** is fixed to the exterior member **50** with fastening members **110**, **111**, and **112**.

[0029] With the above configuration, the rotary operating member **20**, the receiving plate **60**, and the reflecting plate **70** are rotatable together, but their positions in the direction of the rotation axis are fixed. In Present Embodiment, the storage portions **51** and **52** are integrally formed with the exterior member **50**, but there is no limitation to this configuration; the storage portions **51** and **52** formed separately may be fixed to the exterior member **50**.

[0030] In addition, the configuration is not limited to the fixing member **100** being directly fixed to the exterior member **50**, and the fixing member **100** may also be indirectly fixed to the exterior member **50** via another member. Since the storage portions **51** and **52** of Present Embodiment are part of the exterior member **50**, the outside of the exterior member **50** including the storage portions **51** and **52** is defined as the “exterior side”, and the inside of the exterior member **50** is defined as the “interior side”.

[0031] Next, the click mechanism of the mode dial unit **2** will be described with reference to FIGS. **2** and **3**. The groove **21** of the rotary operating member **20** faces the click balls **30** and **31** on the exterior side. In the groove **21**, wavy concave and convex portions are formed at regular intervals.

[0032] Specifically, the wavy concave and convex portions are alternating valleys and peaks extending radially from the axis center. The number of phases of the rotary operating member **20** is denoted as “ n ”. The number of valleys and peaks extending radially from the axis center is n , respectively. In other words, the phase to be rotated by to transition to one mode is “ $360/n$ ” degrees.

[0033] The pair of click balls **30** and **31** are positioned in phase with respect to the wavy concave and convex portions of the groove **21** at the same timing in the valley, and are always biased upward by the compression springs **40** and **41** to the rotary operating member **20** against the wavy concave and convex portions.

[0034] Therefore, the rotary operating member **20** stops at the position where the click balls **30** and **31** fit into the valley of the wavy concave and convex portions. When the user operates the rotary operating member **20** to rotate it by an angle corresponding to one unit position, the click balls **30** and **31** come over the peaks of the wavy concave and convex portions and fit into the adjacent valley.

[0035] At this time, a click force is generated in the rotary operating member **20**, providing the user with a clicking sensation corresponding to one click. Therefore, the user can rotate the rotary operating member **20** accurately by the intended number of clicks.

[0036] Next, a phase detection unit will be described with reference to FIGS. **2** to **6**. In the

following, flexible printed substrates (see FIGS. 2 and 3:90) are simply referred to as the “substrate”. A substrate **90** is fixed to the fixing member **100** with an adhesive material **91** such as double-sided adhesive tape, and is electrically connected to the control circuit of the camera **1** (not illustrated).

[0037] Holes **101**, **102**, and **103** in the fixing member **100** correspond to screw holes **54**, **55**, and **56** which are provided in the exterior member **50**, respectively, through which the fastening members **110**, **111**, and **112** are inserted to fasten the fixing member **100** to the exterior member **50**.

[0038] With the above configuration, the position of the substrate **90** in the rotation axis direction is fixed. A predetermined number of detection units **95** are mounted on the surface of the substrate **90**. The detection unit **95** is an electronic component, such as a photo reflector, for example.

[0039] The photo reflecting plate is a non-contact sensor and incorporates a light emitting source and a light receiving sensor. Light emitted from the light emitting source is reflected by the reflecting plate **70**, and the intensity of the reflected light is detected by the light receiving sensor, thereby measuring the reflectance of the reflecting plate **70**. Various sensors capable of measuring the reflection intensity of the reflecting plate can be used, without being limited to photo reflecting plates.

[0040] Referring to FIG. 4, the number of phases of the rotary operating member **20** and the arrangement of the plurality of detection units **95** will be described. FIG. 4A is a diagram illustrating a first shape pattern related to the reflective member. A hole **74** is formed in the center of rotation of the reflecting plate **70**, and the screws **80** are inserted into the hole **74** to fasten it to the receiving plate **60** and the rotary operating member **20**. When viewed from the direction along the rotation axis of the rotary operating member **20**, the reflecting plate **70** has a plurality of patterns as detection target portions on its inner and outer circumference. A first pattern **70a1** on the outer circumference is the detection target portion formed in a circular shape with the bright portion and the dark portion switching every 180 degrees around the rotation axis.

[0041] A second pattern **70a2** on the inner circumference is a detection target portion formed in a circular shape with the bright portion and the dark portion switching every 90 degrees around the rotation axis. Both the bright portion and the dark portion of the first pattern **70a1** and the second pattern **70a2** have a fan shape extending in an arc.

[0042] The angle at which the bright portion and the dark portion switch in the first pattern **70a1** coincides with the angle at which the bright portion and the dark portion switch in the second pattern **70a2**.

[0043] In the patterns **70a1** and **70a2** of the reflecting plate **70**, the bright portions are portions that reflect the light from the detection unit **95** at a level higher than a predetermined reflectance, and the dark portions are portions that reflect the light from the detection unit **95** at a level lower than the predetermined reflectance.

[0044] For example, the bright portion consists of a bare metal plate such as aluminum, and the dark portion consists of a black coating applied to the metal plate. Alternatively, the dark portion consists of black sheet material and the bright portion consists of white printing on the sheet material.

[0045] By setting the pattern of the reflecting plate **70** to the above settings, the reflective members can be common in a case where the number of phases of the rotary operating member **20** is 2, 4, 6, 8, or 12. In the example in FIG. 4A, the first pattern **70a1** is considered to be the outer circumference, but this is not limited thereto.

[0046] FIG. 4B is a diagram illustrating a second shape pattern regarding the reflective member. In this example, a first pattern **70b1**, which switches between the bright portion and the dark portion every 180 degrees, is on the inner periphery side, and a second pattern **70b2**, which switches between the bright portion and the dark portion every 90 degrees, is on the outer periphery side.

[0047] However, since the number of detection units **95** arranged on the inner periphery side can be reduced in the example of FIG. 4A, it is possible to achieve a greater miniaturization compared to

the example of FIG. 4B.

[0048] Referring to FIG. 4C, the arrangement of the plurality of detection units **95a** to **95f** with respect to the reflecting plate **70** will be described. The first to fourth detection units **95a** to **95d** are installed on the outer periphery side of the substrate **90**, opposite the first pattern **70a1**, and are electrically connected to the substrate **90**. The arrangement of the detection units **95a** to **95d** is as follows. [0049] The first detection unit **95a** is installed in one of a plurality of locations that are divided into 12 areas based on the center of the reflecting plate **70**. [0050] The second detection unit **95b** is installed at a position rotated 90 degrees counterclockwise in FIG. 4C from the first detection unit **95a**. [0051] The third detection unit **95c** is installed at a position rotated 60 degrees in the opposite direction (counterclockwise) to the first detection unit **95a** with respect to the second detection unit **95b**. [0052] The fourth detection unit **95d** is installed at a position rotated 60 degrees in the opposite direction (counterclockwise) from the second detection unit **95b** with respect to the third detection unit **95c**.

[0053] Although the arrangement of the plurality of detection units in the counterclockwise direction is illustrated in FIG. 4C, it is not limited to this example. For each detection unit, the clockwise direction may be used as the reference direction.

[0054] The fifth detection unit **95e** and the sixth detection unit **95f** are arranged opposite the second pattern **70a2**. The fifth detection unit **95e** and the sixth detection unit **95f** are installed at an angular distance of 60 degrees from the center of the reflecting plate **70** and at least 30 degrees from the first detection unit **95a**.

[0055] The advantage of the above arrangement of the patterns **70a1** and **70a2** of the reflecting plate **70** and the plurality of detection units **95** is that the same shape can be used in a case where the number of phases of the rotary operating member **20** is 2, 4, 6, 8, or 12. It is possible to achieve commonality of all other components in the rotary operating device, except for the rotary operating member **20**.

[0056] Referring to FIG. 5, the arrangement of the reflecting plate **70** and the detection unit **95** and the detection signal by the detection unit **95** at each phase number are described. In FIG. 5, for convenience of explanation, the range for each angle of 360 degrees divided by the number of each phase is denoted as areas (1) to (12). The black range in the area corresponds to the dark portion in the pattern of the reflecting plate **70**.

[0057] FIG. 5A-1 illustrates the arrangement of the reflecting plate **70** and the detection unit **95** in a case where the number of phases of the rotary operating member **20** is 2. FIG. 5A-2 illustrates signals of the bright portion and the dark portions detected by the detection unit **95a** in a case where the rotary operating member **20** rotates in the direction of the arrow in FIG. 5A-1.

[0058] The area (1) corresponds to the state in FIG. 5A-1. In a case where the number of phases of the rotary operating member **20** is 2, position detection should be performed by any one of the first to fourth detection units **95a** to **95d**.

[0059] FIG. 5A-2 illustrates the signal detected by the detection unit **95a** in a case where the rotary operating member **20** is in the predetermined position. In the state illustrated in FIG. 5A-1, the detection unit **95a** detects the dark portion of the pattern, but when the rotary operating member **20** is rotated 180 degrees in the direction of the arrow, the bright portion of the pattern is detected. In a case where any of the detection units **95b**, **95c**, or **95d** is selected, the relationship between the bright portion and the dark portion is reversed from the above.

[0060] FIG. 5B-1 illustrates the arrangement of the reflecting plate **70** and the detection unit **95** in a case where the number of phases of the rotary operating member **20** is 4. In FIG. 5B-1, the detection unit **95a** detects the dark portions and the detection unit **95b** detects the bright portions.

[0061] FIG. 5B-2 illustrates signals of the bright portion and the dark portion detected by the detection units **95a** and **95b** in a case where the rotary operating member **20** rotates in the direction of the arrow in FIG. 5B-1. The area (1) corresponds to the state in FIG. 5B-1.

[0062] When the reflecting plate **70** is rotated 90 degrees in the direction of the arrow in FIG. 5B-1,

the dark portion of the reflecting plate **70** is positioned opposite the detection unit **95b**, and both detection units **95a** and **95b** output a signal of the dark portion as shown in the area (2).

[0063] When the reflecting plate **70** is further rotated 90 degrees in the direction of the arrow from this state, the bright portion of the reflecting plate **70** is positioned opposite the detection unit **95a** and the dark portion opposite the detection unit **95b**. As shown in the area (3), the detection unit **95a** outputs a signal of the bright portion and the detection unit **95b** outputs a signal of the dark portion.

[0064] When the reflecting plate **70** is further rotated 90 degrees in the direction of the arrow from this state, the bright portion of the reflecting plate **70** is positioned opposite the detection units **95a** and **95b**. As shown in the area (4), both detection units **95a** and **95b** will be in a state where they both output a signal of the bright portion.

[0065] As described above, phase detection using two detection units **95a** and **95b** allows detection of four different states in the areas (1) to (4) by combining the bright portion and the dark portion.

[0066] FIG. 5C-1 illustrates the arrangement of the reflecting plate **70** and the detection unit **95** in a case where the number of phases of the rotary operating member **20** is 6. FIG. 5C-2 illustrates the signals of the bright portion and the dark portion detected by the detection units **95b**, **95c**, and **95d** in a case where the rotary operating member **20** rotates in the direction of the arrow in FIG. 5C-1. The area (1) corresponds to the state in FIG. 5C-1. This will be described in detail with reference to FIG. 6.

[0067] FIG. 6 is a detailed diagram of each phase when the reflecting plate **70** illustrated in FIG. 5C-1 rotates. FIG. 6A corresponds to FIG. 5C-1, and FIGS. 6B-6F illustrate the reflecting plate **70** rotating counterclockwise every 60 degrees, respectively. FIG. 6G corresponds to FIG. 5C-2.

[0068] FIG. 6A illustrates a state where all of the detection units **95b**, **95c**, and **95d** output a signal of the bright portion (see FIG. 6G: the area (1)). From this state, the reflecting plate **70** rotates 60 degrees in the direction of the arrow to the state illustrated in FIG. 6B.

[0069] The dark portion of the reflecting plate **70** is positioned opposite the detection unit **95b**. As shown in the area (2) of FIG. 6G, the detection unit **95b** outputs the signal of the dark portion, and the detection units **95c** and **95d** both output a signal of the bright portion.

[0070] If the reflecting plate **70** is rotated further by 60 degrees in the direction of the arrow from the state illustrated in FIG. 6B, it attains the state illustrated in FIG. 6C. The dark portion of the reflecting plate **70** is positioned opposite the detection units **95b** and **95c**. As shown in the area (3) of FIG. 6G, the detection units **95b** and **95c** both output a signal of a dark portion, and the detection unit **95d** outputs a signal of a bright portion.

[0071] If the reflecting plate **70** is rotated further by 60 degrees in the direction of the arrow from the state illustrated in FIG. 6C, it attains the state illustrated in FIG. 6D. The dark portion of the reflecting plate **70** is positioned opposite the detection units **95b**, **95c**, and **95d**. As shown in the area (4) of FIG. 6G, the detection units **95b**, **95c**, and **95d** all output a signal of a dark portion.

[0072] If the reflecting plate **70** is rotated further by 60 degrees in the direction of the arrow from the state illustrated in FIG. 6D, it attains the state illustrated in FIG. 6E. The dark portion of the reflecting plate **70** is positioned opposite the detection units **95b** and **95c**. As shown in the area (5) of FIG. 6G, the detection unit **95b** outputs a signal of a bright portion, and the detection units **95c** and **95d** both output a signal of a dark portion.

[0073] If the reflecting plate **70** is rotated further by 60 degrees in the direction of the arrow from the state illustrated in FIG. 6E, it attains the state illustrated in FIG. 6F. The dark portion of the reflecting plate **70** is positioned opposite the detection unit **95d**. As shown in the area (6) of FIG. 6G, the detection units **95b** and **95c** both output a signal of a bright portion, and the detection unit **95d** outputs a signal of a dark portion.

[0074] As described above, phase detection using three detection units **95b**, **95c**, and **95d** allows detection of six different states in the areas (1) to (6) by combining the bright portion and the dark portion.

[0075] FIG. 5D-1 illustrates the arrangement of the reflecting plate **70** and the detection unit **95** in a case where the number of phases of the rotary operating member **20** is 8. FIG. 5D-1 illustrates the state where the detection units **95a** and **95e** output a signal of a dark portion and the detection unit **95b** outputs a signal of a bright portion.

[0076] FIG. 5D-2 illustrates the signals of the bright portion and the dark portion detected by the detection units **95a**, **95b**, and **95e** in a case where the rotary operating member **20** rotates in the direction of the arrow in FIG. 5D-1. The area (1) corresponds to the state in FIG. 5D-1.

[0077] The output state of each detection unit is listed below in a case where the reflecting plate **70** rotates by 45 degrees in the direction of the arrow (counterclockwise) with the state in FIG. 5D-1 as a reference. [0078] At the time of rotation of 45 degrees: The dark portion of the reflecting plate **70** is positioned opposite the detection units **95a**, **95b**, and **95e**, and all three detection units output a signal of the dark portion (see the area (2) in FIG. 5D-2). [0079] At the time of rotation of 90 degrees: The dark portion of the reflecting plate **70** is positioned opposite the detection units **95a** and **95b**, and the detection units **95a** and **95b** output the signal of the dark portion and the detection unit **95e** outputs the signal of the bright portion (see the area (3) in FIG. 5D-2). [0080] At the time of rotation of 135 degrees: The dark portion of the reflecting plate **70** is positioned opposite the detection unit **95b**. The detection unit **95b** outputs a signal of a dark portion, and the detection units **95a** and **95e** both output a signal of the bright portion (see the area (4) in FIG. 5D-2). [0081] At the time of rotation of 180 degrees: The dark portion of the reflecting plate **70** is positioned opposite the detection units **95b** and **95e**. The detection units **95b** and **95e** both output a signal of a dark portion and the detection unit **95a** outputs a signal of the bright portion (see the area (5) in FIG. 5D-2). [0082] At the time of rotation of 225 degrees: The dark portion of the reflecting plate **70** is positioned opposite the detection unit **95e**. The detection unit **95e** outputs a signal of a dark portion, and the detection units **95a** and **95b** both output a signal of a bright portion (see the area (6) in FIG. 5D-2). [0083] At the time of rotation of 270 degrees: The bright portion of the reflecting plate **70** is positioned opposite the detection units **95a**, **95b**, and **95e**. All three detection units output a signal of a bright portion (see the area (7) in FIG. 5D-2). [0084] At the time of rotation of 315 degrees: The dark portion of the reflecting plate **70** is positioned opposite the detection unit **95a**. The detection unit **95a** outputs a signal of a dark portion, and the detection units **95b** and **95e** both output a signal of a bright portion (see the area (8) in FIG. 5D-2).

[0085] As described above, phase detection using three detection units **95a**, **95b**, and **95e** allows detection of eight different states in the areas (1) to (8) by combining the bright portion and the dark portion.

[0086] FIG. 5E-1 illustrates the arrangement of the reflecting plate **70** and the detection unit **95** in a case where the number of phases of the rotary operating member **20** is 12. FIG. 5E-1 illustrates the state where the detection units **95a** and **95e** output a signal of a dark portion and the detection units **95b** and **95f** output a signal of a bright portion.

[0087] FIG. 5E-2 illustrates signals of a bright portion and a dark portion detected by the detection units **95a**, **95b**, **95e**, and **95f** in a case where the rotary operating member **20** rotates in the direction of the arrow in FIG. 5E-1.

[0088] The area (1) corresponds to the state in FIG. 5E-1. The output state of each detection unit is listed below in a case where the reflecting plate **70** rotates by 30 degrees in the direction of the arrow (counterclockwise) with the state in FIG. 5E-1 as a reference. [0089] At the time of rotation of 30 degrees: The dark portion of the reflecting plate **70** is positioned opposite the detection units **95a**, **95b**, and **95e**. The detection units **95a**, **95b**, and **95e** all output a signal of a dark portion, and the detection unit **95f** outputs a signal of a bright portion (see the area (2) in FIG. 5E-2). [0090] At the time of rotation of 60 degrees: The dark portion of the reflecting plate **70** is positioned opposite the detection units **95a**, **95b**, **95e**, and **95f**. All four detection units will be in a state where they all output a signal of a dark portion (see the area (3) in FIG. 5E-2). [0091] At the time of rotation of 90 degrees: The dark portion of the reflecting plate **70** is positioned opposite the detection units **95a**,

95b, and **95f**. The detection units **95a**, **95b**, and **95f** all output a signal of a dark portion, and the detection unit **95e** outputs a signal of a bright portion (see the area (4) in FIG. 5E-2). [0092] At the time of rotation of 120 degrees: The dark portion of the reflecting plate **70** is positioned opposite the detection units **95b** and **95f**. The detection units **95b** and **95f** both output a signal of a dark portion and the detection units **95a** and **95e** both output a signal of a bright portion (see the area (5) in FIG. 5E-2). [0093] At the time of rotation of 150 degrees: The dark portion of the reflecting plate **70** is positioned opposite the detection unit **95b**. The detection unit **95b** outputs a signal of a dark portion, and the detection units **95a**, **95e**, and **95f** all output a signal of a bright portion (see the area (6) in FIG. 5E-2). [0094] At the time of rotation of 180 degrees: The dark portion of the reflecting plate **70** is positioned opposite the detection units **95b** and **95e**. The detection units **95b** and **95e** both output a signal of a dark portion and the detection units **95a** and **95f** both output a signal of a bright portion (see the area (7) in FIG. 5E-2). [0095] At the time of rotation of 210 degrees: The dark portion of the reflecting plate **70** is positioned opposite the detection unit **95e**. The detection unit **95e** outputs a signal of a dark portion, and the detection units **95a**, **95b**, and **95f** all output a signal of a bright portion (see the area (8) in FIG. 5E-2). [0096] At the time of rotation of 240 degrees: The dark portion of the reflecting plate **70** is positioned opposite the detection units **95e** and **95f**. The detection units **95e** and **95f** both output a signal of a dark portion and the detection units **95a** and **95b** both output a signal of a bright portion (see the area (9) in FIG. 5E-2). [0097] At the time of rotation of 270 degrees: The dark portion of the reflecting plate **70** is positioned opposite the detection unit **95f**. The detection unit **95f** outputs a signal of a dark portion, and the detection units **95a**, **95b**, and **95e** all output a signal of a bright portion (see the area (10) in FIG. 5E-2). [0098] At the time of rotation of 300 degrees: The dark portion of the reflecting plate **70** is positioned opposite the detection units **95a** and **95f**. The detection units **95a** and **95f** both output a signal of a dark portion, and the detection units **95b** and **95e** both output a signal of a bright portion (see the area (11) in FIG. 5E-2). [0099] At the time of rotation of 330 degrees: The dark portion of the reflecting plate **70** is positioned opposite the detection unit **95a**. The detection unit **95a** outputs a signal of a dark portion, and the detection units **95b**, **95e**, and **95f** all output a signal of a bright portion (see the area (12) in FIG. 5E-2).

[0100] As described above, phase detection using four detection units **95a**, **95b**, **95e**, and **95f** allows detection of twelve different states in the areas (1) to (12) by combining the bright portion and the dark portion.

[0101] Although FIG. 5 illustrates the arrangement of the detection units and their output signals at each phase number, the number of the detection units **95** mounted on the substrate may be all six, as illustrated in FIG. 4C.

[0102] Alternatively, as illustrated in FIGS. 5A-1, 5B-1, 5C-1, 5D-1, and 5E-1, depending on the number of phases, the number of detection units on the board can be reduced and contribute to cost reduction.

[0103] Next, a method for aligning the phase between the rotary operating member **20** and the reflecting plate **70** will be described. The rotary operating member **20** has the groove **21** extending radially from the shaft center, and is formed of wavy concave and convex portions with alternating valleys and peaks extending radially from the shaft center at regular intervals.

[0104] The apexes of the mountains coincide in rotational phase with the switching position of the bright portion and the dark portion of the first pattern **70a1** or the second pattern **70a2** of the reflecting plate **70**. If the number of phases is expressed as n , the bottom position of the groove **21** corresponds to the position rotated by " $180/n$ " degrees from the apex of the mountain above.

[0105] For example, in a case where the number of phases illustrated in FIG. 5E-1 is 12, the click balls **30** and **31** engage the groove **21** every 30 degree phase. At that time, the reflecting plate **70** stays in the position corresponding to one of the areas (1) to (12).

[0106] Since the click balls **30** and **31** reliably stop in a state corresponding to any one of the areas (1) to (12), the user can sense the click force, and the detection unit **95** can stably detect a signal.

[0107] In the above example, a click mechanism using the groove **21** of the rotary operating member **20**, the click balls **30** and **31**, and the compression springs **40** and **41** is described, but is not limited to this configuration. A modification embodiment is illustrated in FIGS. **7** and **8**.

[0108] FIG. **7** illustrates an example of a retention mechanism for a rotary operating member as the first modification embodiment. FIG. **7A** is an exploded perspective view of the rotary operating member. FIG. **7B** is a perspective view of a receiving plate **260** having the groove **261**. The difference is that in the example of FIG. **2**, the groove **21** is formed on the rotary operating member **20**, but in the example of FIG. **7**, a groove **261** is formed on the receiving plate **260**.

[0109] In the first modification embodiment, the click balls **230** and **231** are biased against the grooves **261** by compression springs **240** and **241**, thereby holding the rotary operating member **220**. In addition, the direction of force of the click balls **230** and **231** is downward in FIG. **7**, but it can be upward. For example, the components of the receiving plate **260** may be arranged inside the rotary operating member **220**.

[0110] Referring to FIG. **8**, another example of the retention mechanism for the rotary operating member is shown as a second modification embodiment. FIG. **8** illustrates a bottom view of the mode dial unit **2** as seen from the inside. A groove **361** is formed in a separate component from the rotary operating member, and the click ball **330** is biased by a compression spring **340** to engage with the groove **361**.

[0111] However, the direction of force of the click ball **330** is radial, and the concave portion and the convex portion in the groove **361** are formed continuously along the circumferential direction. For example, the groove **361** can be formed on reflecting plate **70** illustrated in FIG. **2**.

[0112] Forming the groove **361** in the reflecting plate **70** reduces the number of components. Similar to the examples in FIG. **2** and FIG. **7**, the configuration in FIG. **8** can also be used to maintain the rotation phase of the rotary operating member **20** and improve the operating feel due to the click force. In particular, in a case where the vertical dimension of the rotary operating member is to be reduced, it is desirable to adopt the configuration shown in the second modification embodiment.

[0113] According to the above embodiments and modification embodiments, the optical detection method reduces false detection, the number of phases (resolution) can be easily changed, the rotation of the rotary operating member over the entire circumference is possible, and a more compact rotary operating device can be provided.

[0114] Although the preferred embodiment of the present invention has been described above, the present invention is not limited to the above embodiment, and various modifications and changes are possible within the scope of the gist of the present invention. The electronic device in which the rotary operating member is mounted is not limited to imaging devices, but can be applied to a wide range of household electrical appliances, measuring instruments, vehicles, or the like.

[0115] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation to encompass all such modifications and equivalent structures and functions.

[0116] This application claims the benefit of priority from Japanese Patent Application No. 2024-034149, filed on Mar. 6, 2024, which is hereby incorporated by reference herein in its entirety.

Claims

1. A rotary operating device comprising: a rotary operating member that can rotate around a rotary axis; a reflective member that can rotate together with the rotary operating member; and a detection unit configured to detect reflected light from the reflective member, wherein the reflective member is provided with a plurality of detection target portions each having a bright portion and a dark portion, a first detection target portion of the plurality of detection target portions is a portion

formed in an annular shape in which a bright portion and a dark portion switch every 180 degrees around the rotation axis, a second detection target portion of the plurality of detection target portions is a portion formed in an annular shape in which a bright portion and a dark portion switch every 90 degrees around the rotation axis, and an angle at which the bright portion and dark portion switch at the first detection target portion is the same as the angle at which the bright portion and dark portion switch at the second detection target portion.

2. The rotary operating device according to claim 1, wherein the first detection target portion is positioned on an outer circumference of the reflective member and the second detection target portion is positioned on an inner circumference of the reflective member.

3. The rotary operating device according to claim 1, wherein the first detection target portion is positioned on the inner circumference of the reflective member and the second detection target portion is positioned on the outer circumference of the reflective member.

4. The rotary operating device according to claim 1, wherein, the detection unit is configured to oppose the first detection target portion.

5. The rotary operating device according to claim 1, wherein the detection unit includes a first detection unit and a second detection unit both configured to oppose the first detection target portion, wherein the first and second detection units are installed at intervals of 90 degrees around the rotation axis.

6. The rotary operating device according to claim 1, wherein the detection unit includes a first detection unit to a third detection unit which are configured to oppose the first detection target portion, wherein the first to third detection units are installed at intervals of 60 degrees around the rotation axis.

7. The rotary operating device according to claim 5, wherein the detection unit includes a third detection unit configured to oppose the second detection target portion.

8. The rotary operating device according to claim 5, wherein the detection unit includes a third detection unit and a fourth detection unit both configured to oppose the second detection target portion, wherein the third and fourth detection units are installed at intervals of 60 degrees around the rotation axis and at least 30 degrees apart from the first detection unit around the rotation axis.

9. The rotary operating device according to claim 8, wherein the third detection unit is installed on the opposite side of the first detection unit with respect to the position of the fourth detection unit.

10. The rotary operating device according to claim 1, further comprising a retention mechanism for the rotary operating member, wherein the retention mechanism has a sphere, and an elastic member, and an elastic member forces the sphere into a groove provided around the rotary operating member.

11. The rotary operating device according to claim 10, wherein the groove is provided in the rotary operating member.

12. The rotary operating device according to claim 10, wherein the groove is provided on a receiving plate of the reflective member.

13. The rotary operating device according to claim 10, wherein the reflective member has the groove.

14. The rotary operating device according to claim 1, wherein the number of detection units is different depending on the number of phases of the rotary operating member.

15. An electronic device comprising: the rotary operating device according to claim 1.
